Algebraic Torsion in Minimal Einstein–Cartan–Holst Gravity: Four-Fermion Contact Bounds and Classical Scalar-Sector Transparency

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We isolate two consequences of the minimal Einstein–Cartan–Holst (ECH) action that can be established without a phenomenological dark-energy map. First, varying the tetrad, Lorentz connection, and minimally coupled Dirac field in first order and eliminating the non-propagating torsion gives the standard axial–axial four-fermion contact interaction with coefficient of order $M_{\rm Pl}^{-2}$. At the explicitly evaluated homogeneous benchmark $n_\psi = 100\,{\rm cm}^{-3}$, its coefficient-one scale κn_ψ^2 is $3.5571 \times 10^{-69}\rho_\Lambda$. This number is a finite-density scale comparison, not a derived vacuum stress tensor or equation of state. Within a hard-four-momentum-cutoff, direct-channel, standard mean-field Nambu–Jona-Lasinio (NJL) treatment, the scalar Fierz projection is repulsive, $G_{\rm scalar} = -3\kappa/64$, so the declared scalar gap equation has no nonzero solution. This is a sign result, not a blanket magnitude result: after enforcing the unreduced-Planck convention and the normalization of $G_s(\bar\psi\psi)^2$, the scalar coefficient-to-threshold ratio reaches 1.96039 at the formal above-Planck stress-test point, and the axial coefficient benchmark is twice the scalar ratio. These statements constrain this specific contact channel; they do not exclude beyond-mean-field dynamics, non-minimal fermion couplings, additional species structure, propagating torsion, or the full gravitational effective field theory.

Second, for minimally coupled canonical scalar matter the spin current vanishes, so the algebraic Cartan equation selects the torsion-free branch. On that branch the Holst contraction vanishes pointwise by the first algebraic Bianchi identity. Consequently the constant-Immirzi Holst sector does not modify classical scalar or tensor perturbation equations at any order about that branch. This is a narrow classical transparency statement, not a claim about fermion loops, anomalies, a dynamical Immirzi field, non-minimal matter, or propagating torsion.

We deliberately make no ECH dark-energy or birefringence prediction. The one-loop charge flow studied by Shapiro and Teixeira and the Immirzi running studied by Benedetti and Speziale do not by themselves supply a derived Lorentzian cosmological stress tensor or observable map; those Route-2/3 identifications therefore remain unresolved literature context rather than closures in this paper. Generic stock-CAMB, NaMaster, and spectator-ALP consistency studies are reported separately in Paper I(b) and are not evidence for ECH.

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CONTENTS

I. INTRODUCTION

In Einstein–Cartan gravity the Lorentz connection is independent of the tetrad, and fermionic spin sources torsion. For the minimal action the connection equation is algebraic: torsion does not propagate and may be eliminated exactly, leaving a local four-fermion interaction [1–3]. The Holst extension introduces the dimensionless Barbero–Immirzi parameter but no new propagating de-

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gree of freedom when that parameter is constant. These elementary facts permit two useful, auditable questions: how large can the induced contact channel be at late times, and can the constant-Immirzi Holst term affect classical perturbations sourced only by canonical scalars?

This paper answers those two questions within an intentionally narrow domain. Section II states the first-order action and performs the algebraic torsion elimination. Section III A evaluates the induced contact channel and a standard mean-field NJL gap equation. The Fierz rearrangement used in that calculation is recorded in Appendix A, and the regulated gap-equation audit is in Appendix B. Section V gives the exact classical scalar-sector transparency argument.

The claim boundary is as important as the calculation. A Fierz identity for the contact interaction does not enumerate the gravitational EFT. A mean-field NJL calculation does not eliminate Fierz ambiguity or beyond-mean-field strong-coupling completions. Likewise, vanishing of the Holst density on the torsion-free scalar branch is not a theorem about fermionic, quantum, non-minimal, dynamical-Immirzi, or propagating-torsion sectors. Earlier exploratory mappings from one-loop Holst/Immirzi running to late-time dark energy are not retained as results: the cited calculations do not derive the required cosmological stress tensor and observable matching [4, 5]. No operator-complete or unrestricted no-go is claimed.

Paper I(b) [6] is a separate computational companion. Its stock-CAMB extra-radiation chains, synthetic NaMaster recovery test, and generic spectator-ALP fit are proxy, pipeline, and consistency studies. They are not inputs to either result proved here and are not evidence for ECH.

II. MINIMAL ECH AND ALGEBRAIC TORSION

We use the first-order Einstein–Cartan–Holst action with a constant Barbero–Immirzi parameter γ and a minimally coupled Dirac field,

$$S[e,\omega,\psi] = \frac{1}{4\kappa}\int\left(\epsilon_{IJKL} + \frac{2}{\gamma}\eta_{I[K}\eta_{L]J}\right) \times e^I\wedge e^J\wedge R^{KL}(\omega) + S_{\mathrm{D}}[e,\omega,\psi], \tag{1}$$

where $\kappa = 8\pi G = 8\pi/M_{\mathrm{Pl}}^2 = 1/\bar{M}_{\mathrm{Pl}}^2$, $M_{\mathrm{Pl}} \equiv G^{-1/2} = 1.22089\times10^{19}$ GeV is the unreduced Planck mass, $\bar{M}_{\mathrm{Pl}} \equiv (8\pi G)^{-1/2}$ is the reduced Planck mass, $R^{IJ}(\omega) = d\omega^{IJ} + \omega^I{}_K\wedge\omega^{KJ}$, and e^I and ω^{IJ} are varied independently. There is no independent T^2 term in Eq. (1); any torsion-square or four-fermion expression below is an on-shell result obtained only after the connection equation has been solved. This ordering avoids double counting.

For minimal fermion coupling the spin current is totally antisymmetric and dual to $J_5^I = \bar\psi\gamma^I\gamma^5\psi$. The connection equation is algebraic. In the Einstein–Cartan limit, with the spin-current convention $S^{IJK} = \frac{1}{4}\epsilon^{IJKL}J_{5L}$ used here, it reads

$$T^{IJK} = \kappa S^{IJK} = \frac{\kappa}{4}\epsilon^{IJKL}J_{5L}. \tag{2}$$

At finite γ the algebraic solution acquires the standard Holst contorsion prefactor; solving it and substituting it back into the action gives the minimal axial–axial contact interaction [3, 7],

$$\mathcal{L}_{4\psi} = -\frac{3\kappa}{16}\frac{\gamma^2}{1+\gamma^2}\left(J_5^I J_{5I}\right). \tag{3}$$

The pure Einstein–Cartan coefficient is recovered for $\gamma\to\infty$. A vector–axial term is not generated by the minimal coupling used here; it requires a non-minimal fermion coupling and is outside this paper's declared field content.

For the conservative late-time and gap-equation bounds below we use the maximal Einstein–Cartan magnitude,

$$\mathcal{L}_{\mathrm{tor}}^{\mathrm{NJL}} = -\frac{3\kappa}{16}(J_5^I J_{5I}), \tag{4}$$

because $\gamma^2/(1+\gamma^2) < 1$ can only reduce the coupling. Equations (1)–(4) are the complete action-level input to the results that follow.

III. THE MINIMAL FOUR-FERMION CONTACT CHANNEL

The contact operator in Eq. (5) is a specific consequence of minimal algebraic torsion. We test its finite-density amplitude and its standard mean-field scalar-condensate channel. Neither test is promoted to a claim about every four-fermion completion or about the complete gravitational EFT.

A. Finite-density benchmark and equation-of-state boundary

On the standard Einstein–Cartan side—i.e. before the Holst term is added—the Cartan algebraic equation $T^{abc} = \kappa\, S^{abc}$ (Eq. (2); in the half-weight torsion convention $T^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{[\mu\nu]}$ of the original literature this reads $T^{abc} = (\kappa/2)\,\bar\psi\gamma^{[a}\gamma^b\gamma^{c]}\psi$ — the two map exactly, see the convention footnote at Eq. (2)) allows torsion to be integrated out exactly, generating an effective four-fermion contact term whose coefficient is the gravitational coupling $\kappa = 8\pi G$ [1, 2]. Following the standard Hehl–Datta derivation, the resulting axial–axial contact interaction is

$$\mathcal{L}_{\mathrm{tor}}^{\mathrm{NJL}} = -\frac{3}{16}\,\kappa\left(\bar\psi\,\gamma^a\gamma^5\,\psi\right)^2, \tag{5}$$

i.e. a four-fermion operator suppressed by M_{Pl}^{-2} and *parity-even* in the *CP*-conserving Standard Model sector.

The coefficient scale for a specified homogeneous finite-density state obeys the conservative benchmark $|\rho_{4f}| \lesssim \kappa n_\psi^2 = 8\pi n_\psi^2/M_{\rm Pl}^2$. The actual maximal axial–axial coefficient in Eq. (5) adds the factor 3/16. Since n_ψ has mass dimension +3, both expressions have the required energy-density dimension +4. We evaluate this expression at the deliberately elevated homogeneous benchmark $n_\psi = 100\,\mathrm{cm}^{-3}$ — not the cosmic-mean baryon density (which is $\sim 2\times 10^{-7}\,\mathrm{cm}^{-3}$ today and would make the bound stronger still) — converted to natural units via $\hbar c = 1.973269804\times 10^{-5}\,\mathrm{eV\,cm}$ ($1\,\mathrm{cm}^{-3} = (1.973269804\times 10^{-5}\,\mathrm{eV})^3 \approx 7.68351\times 10^{-15}\,\mathrm{eV}^3$, so $n_\psi \approx 7.68351\times 10^{-13}\,\mathrm{eV}^3$), and with $M_{\rm Pl} = 1.22089\times 10^{28}\,\mathrm{eV}$, gives

$$\kappa n_\psi^2 \approx 9.9542\times 10^{-80}\,\mathrm{eV}^4, \quad \frac{\kappa n_\psi^2}{\rho_\Lambda} \approx 3.5571\times 10^{-69}. \tag{6}$$

This conservative bound is 68.45 orders below the canonical $\rho_\Lambda \approx (2.3\,\mathrm{meV})^4 \approx 2.8\times 10^{-11}\,\mathrm{eV}^4$ used throughout this paper. Including the actual 3/16 contact coefficient gives $1.8664\times 10^{-80}\,\mathrm{eV}^4 = 6.6695\times 10^{-70}\rho_\Lambda$. No coherent order parameter, vacuum expectation, stress tensor, or equation of state is inferred from $\langle J^5\rangle \approx 0$: a one-point current does not determine the state-dependent composite $\langle J^5 J^5\rangle$ or its metric variation. The only late-density statement made here is the explicit finite-density coefficient-scale comparison above. Thus the coefficient-independent finite-density bound is 68.45 orders below ρ_Λ even under the elevated homogeneous-density choice, while the cosmic-mean density gives a stronger bound. This is a constraint on the minimal Hehl–Datta contact channel, not a claim that arbitrary torsion or fermion sectors cannot contribute to late-time cosmology. The finite-γ factor in Eq. (3) is bounded by unity and cannot enhance this specific channel above the Einstein–Cartan limit used in the estimate.

B. Standard mean-field NJL check

The finite-density benchmark does not decide whether a vacuum condensate exists. As a logically separate conditional check, we Fierz-project the axial–axial operator and solve a hard-cutoff NJL gap equation in Appendix B. In the declared direct-channel convention the scalar projection is $G_{\rm scalar} = -3\kappa/64 < 0$, hence repulsive. Substitution in the real homogeneous scalar gap equation shows that this negative coupling has no nonzero solution. The sign conclusion is logically separate from the coefficient-magnitude diagnostic: the latter is not subcritical throughout the scan, reaching $|G_{\rm scalar}|/G_{\rm crit} = 1.96039$ for $N_f N_c = 9$ at the formal $\Lambda = M_{\rm Pl}/\sqrt{0.274}$ stress point. The formal axial coefficient divided by the same scalar-channel threshold is twice the scalar ratio; it is not an independently derived axial-condensation threshold. The full six-row scan is reported in Appendix B. The machine-checkable calculation is `njl_gap_equation_route1.py`.

This result is deliberately conditional. It closes the scalar condensate in the standard mean-field NJL model with the stated regulator, field content, and projection convention. It does not remove the known Fierz ambiguity of mean-field truncations and does not exclude beyond-mean-field strong coupling, additional flavor/color exchange structure, or non-minimal torsion couplings.

IV. RELATION TO EARLIER RUNNING CALCULATIONS

Two primary-source results motivate, but do not complete, possible extensions of the minimal calculation. Shapiro and Teixeira studied one-loop renormalization of Einstein–Cartan gravity with the Holst term and external currents; their coupled charge system has no finite-Immirzi fixed point and was not solved into a unique infrared normalization [4]. Benedetti and Speziale derived an Immirzi beta function in a specified fermionic truncation, with explicit scheme and perturbative limitations [5, 8]. Neither calculation derives a Lorentzian cosmological stress tensor or an observable late-time amplitude from the running alone. Consequently this paper does not identify either result with dark energy, birefringence, or a closed cosmological channel. Such a claim would require an explicit matching calculation that is not presently available.

V. CLASSICAL SCALAR-SECTOR TRANSPARENCY

A. Statement

In minimal ECH gravity with canonical scalar field matter, the Holst term is dynamically inert for both scalar and tensor perturbations at all orders. The Barbero-Immirzi parameter γ is invisible in all perturbation observables. This generalizes Hehl *et al.* (1976) [1] to the Holst sector and to all perturbation orders. The restriction to the torsion-free ($T = 0$) branch is not an additional assumption but a *consequence* of the matter content: canonical scalar matter carries zero spin density (Step 1 below), hence in Einstein–Cartan theory sources no torsion (Step 2), and the connection reduces to Levi-Civita identically at all perturbation orders. We emphasize that the "all orders" statement does *not* rest on an order-by-order component expansion of the linearized Holst/Cartan action: it follows because Step 2 is an *algebraic* (non-derivative) Cartan constraint, so $T = 0$ holds exactly and perturbatively unmodified, and Step 4 is a pointwise *identity* (the algebraic Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$, which holds for any torsion-free connection at every field configuration). Because both load-bearing steps are exact identities rather than truncated expansions, no separate order-by-order verification is required; we display the leading (second-order) expansion in Sec. V D only as an explicit check of the identity, not as the origin of the all-orders claim. The propagating-torsion, dynamical-Immirzi-field, fermion-loop, and non-minimal-matter sectors — where $T \neq 0$ and the result need not hold — are explicitly outside the stated scope; the theorem is a statement about the scalar-matter sector of ECH, which is precisely the Levi-Civita branch, so the $T = 0$ hypothesis is the theorem's domain rather than a gap in it.

B. Proof (Scalar Sector)

1. **Zero spin density.** A canonical scalar field has zero spin density.

2. **Zero torsion.** In Einstein-Cartan theory, $T^\lambda{}_{\mu\nu} = 8\pi G\, S^\lambda{}_{\mu\nu} + \ldots$. With $S = 0$, $T^\lambda{}_{\mu\nu} = 0$ at all perturbation orders.

3. **Connection reduces to Levi-Civita.** $\Gamma^\lambda{}_{\mu\nu} = \mathring{\Gamma}^\lambda{}_{\mu\nu}$.

4. **Holst term vanishes by the first Bianchi identity.** The Holst term evaluated with the Levi-Civita connection gives $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\mathring{\Gamma})$, which on a torsion-free connection *vanishes identically* by the first (algebraic) Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$: the cyclic-sum identity $R_{\mu\nu\rho\sigma} + R_{\mu\rho\sigma\nu} + R_{\mu\sigma\nu\rho} = 0$ contracted with the totally antisymmetric $\epsilon^{\mu\nu\rho\sigma}$ leaves no non-trivial component. The algebraic Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$ holds for any torsionless connection, independently of metric compatibility; non-metricity does not invalidate the identity provided $T = 0$. The Holst dual contraction is therefore identically zero on the Levi-Civita connection (not merely a boundary term), so it contributes nothing to the action at any order. This Bianchi-vanishing is distinct from the Pontryagin density $\propto R\,\widetilde{R}$ (a two-curvature topological invariant) — see the explicit verification below.

5. **No equations of motion.** A total derivative contributes nothing to variational equations at all orders. (This step is logically distinct from the pointwise Bianchi-vanishing of the previous step: at $T = 0$ the stronger pointwise vanishing already applies; the total-derivative statement covers the residual Nieh–Yan boundary term $d(e_I \wedge T^I)$ at nonzero torsion.)

C. Extension to Tensor Sector

The same five steps apply to tensor perturbations. With $T = 0$, the tensor perturbation equation:

$$h''_{ij} + 2\mathcal{H}h'_{ij} + k^2 h_{ij} = 0 \tag{7}$$

(primes denote derivatives with respect to conformal time η, and $\mathcal{H} \equiv a'/a$ is the conformal Hubble rate; Fourier modes $h_{ij}(\mathbf{k},\eta)$ follow the standard $e^{i\mathbf{k}\cdot\mathbf{x}}$ convention for the transverse-traceless amplitude; in cosmic time, using $dt = a\,d\eta$, the equivalent form is $\ddot{h}_{ij} + 3H\dot{h}_{ij} + (k^2/a^2)h_{ij} = 0$) has no parity-dependent modifications. Left and right circular polarization modes propagate identically:

$$v_R(k,\eta) = v_L(k,\eta) \quad \Rightarrow \quad \Delta v = 0 \quad \text{(identically).} \tag{8}$$

No GW birefringence, no tensor chirality, no TB/EB CMB parity violation from the ECH mechanism.

D. Explicit Verification: The Holst Term in Perturbation Theory

Expanding to second order, the Holst dual evaluates on the Levi-Civita connection ($T = 0$) as:

$$\mathcal{R}_{\rm H}(\mathring{\Gamma}) \equiv \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\mathring{\Gamma}) \;=\; 0 \quad \text{(identically, by the first (algebraic) Bianchi identity).} \tag{9}$$

This is the Bianchi-vanishing of the Holst dual contraction on a torsion-free connection: in differential-form language $e^I \wedge e^J \wedge R_{IJ} = -\mathrm{NY} + T^I \wedge T_I$, where $\mathrm{NY} \equiv d(e_I \wedge T^I)$ is the exact Nieh–Yan density (a total derivative of the torsion boundary term), and both pieces vanish at $T = 0$ *pointwise* — not merely up to a boundary term: $\mathrm{NY}|_{T=0} = d(0) = 0$ and $T^I \wedge T_I|_{T=0} = 0$, with the Bianchi cancellation above being the operative reason the Holst dual vanishes pointwise in the torsionless sector.[1] *This must be carefully distinguished from the Pontryagin density* $\frac{1}{4}\,\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu}{}^{\alpha\beta}R_{\rho\sigma\alpha\beta} \propto R\,\widetilde{R}$, which involves *two*

[1] Explicit decomposition: the Holst integrand satisfies $e^I \wedge e^J \wedge R_{IJ}(\Gamma) = -d(e_I \wedge T^I) + T^I \wedge T_I$, i.e. $e \wedge e \wedge R = -\mathrm{NY} + T \wedge T$, where $\mathrm{NY} \equiv d(e_I \wedge T^I)$ is the Nieh–Yan boundary form and $T^I = de^I + \omega^I{}_J \wedge e^J$ is the torsion two-form. At $T = 0$ (canonical scalar matter, torsion-free branch), both terms vanish *pointwise*: $\mathrm{NY}|_{T=0} = d(0) = 0$ and $T^I \wedge T_I|_{T=0} = 0$. The Holst dual contraction is therefore identically zero on the Levi-Civita connection, not merely a total derivative.

curvature tensors and is a separate true topological invariant — non-zero pointwise and a total derivative even in the presence of torsion. The Holst dual contraction has only *one* curvature and is therefore not the Pontryagin density; on $T = 0$ it is identically zero by Bianchi, contributing nothing to the variational equations of motion at any perturbation order. In particular, the cubic action for ζ (which determines the bispectrum) receives zero contribution from the Holst term. The bispectrum is therefore identical to the standard GR result.

E. What Would Break the Transparency

The transparency result fails if: (1) matter includes fermions with nonzero spin density; (2) the gravitational action includes kinetic terms for torsion (Poincaré gauge theory); (3) non-minimal derivative couplings between the Holst sector and matter are introduced.

Quantum scope limitation.—The perturbation-transparency result is established at the *classical* level: it follows from the algebraic Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$ on the torsion-free Levi-Civita connection, which is an exact classical identity, and from the algebraic (non-derivative) Cartan constraint that sources no torsion for canonical scalar matter. Whether classical Holst-sector decoupling survives quantum loop corrections—in particular, whether graviton loops, matter loops, or chiral/gravitational anomalies could reintroduce propagating torsion degrees of freedom or source an effective Holst contribution at the quantum level—is *not* claimed here and lies outside the stated scope of this result. The perturbation-transparency theorem is a statement about the classical scalar-matter sector of ECH; its quantum extension would require a separate analysis of the one-loop effective action in the Holst+scalar sector.

F. Implications

The perturbation-transparency result establishes a clean dichotomy:

- *Perturbation observables* (C_ℓ^{TT}, C_ℓ^{EE}, P_k, bispectrum): Identical to standard GR. No ECH modifications at any order.
- *Nonperturbative parity channels* (ALP birefringence, primordial GW chirality): Parity-sensitive channels (model-dependent tests of γ_{BI} only under a derived γ_{BI}-dependent photon or tensor-parity coupling).

VI. DISCUSSION AND LIMITATIONS

The two results have different logical status. Algebraic torsion elimination is an action-level calculation. The finite-density estimate then follows from dimensional scaling of that derived contact operator, whereas the NJL statement additionally assumes a specific mean-field reduction and regulator. The transparency result is exact within its classical domain because both load-bearing steps—zero scalar spin current and the algebraic Bianchi identity—hold pointwise. Its novelty is not a new curvature identity; its utility is to make the observational consequence and claim boundary explicit for the constant-Immirzi, canonical-scalar branch.

Several stronger conclusions are not supported. This paper does not enumerate all diffeomorphism-invariant operators, establish vacuum stability beyond the stated NJL truncation, or constrain non-minimal fermion couplings. It does not analyze propagating torsion, a dynamical Immirzi field, quantum loops or anomalies in the scalar sector, or boundary/topological sectors with nontrivial global data. Most importantly, it does not derive a late-time vacuum energy, cosmic birefringence, or a bounce observable from ECH. These are residual research problems, not closures hidden by terminology.

VII. CONCLUSIONS

Minimal ECH gravity supplies a definite, Planck-suppressed axial contact interaction after algebraic torsion is integrated out. At the stated $n_\psi = 100\,\mathrm{cm}^{-3}$ homogeneous benchmark its coefficient-one scale is $3.5571 \times 10^{-69}\rho_\Lambda$; this is not an equation-of-state or vacuum-stress calculation. Separately, its scalar projection has no nonzero real homogeneous solution in the declared direct-channel, hard-four-momentum- cutoff, standard mean-field gap equation because its coupling is repulsive. This conditional conclusion does not use a blanket magnitude-subcriticality claim: the formal $N_f N_c = 9$, above-Planck stress point is magnitude-supercritical. Separately, canonical scalar matter sources no torsion, and the constant-Immirzi Holst term vanishes on the resulting torsion-free branch by the first Bianchi identity, leaving classical scalar and tensor perturbation equations unchanged.

These are the complete claims of this paper. The evaluated finite-density benchmark is far below the observed dark-energy scale, but no conclusion about a coherent mean field, vacuum stress tensor, or equation of state is drawn from $\langle J^5 \rangle \approx 0$. Independently, the classical Holst sector leaves scalar/tensor perturbations unchanged on the stated torsion-free scalar branch; it does not yield a bounce observable there. This does not constitute an operator-complete no-go or an ECH explanation of dark energy. The unresolved step in proposed running-based extensions is the derivation of a matched Lorentzian cosmological stress tensor and observable from the underlying one-loop/RG calculation. Until that step exists, the honest conclusion is a specific contact-channel bound and a narrow classical transparency identity.

Data and Code Availability

The algebraic checks used here are committed at `fierz_lemma_check.py` and `njl_gap_equation_route1.py`; the latter writes the exact six-row and density record `njl_gap_equation_route1_results.json`. The repository and version history are available at https://github.com/Hubify-Projects/bigbounce. Paper I(b) is a separate submission with its own computational artifacts; none of them is required to reproduce the action-level results in this paper.

Appendix A: Fierz Rearrangement of the Minimal Axial Contact Operator

On the 16-dimensional Clifford basis, grouped into the five Lorentz classes $(S, V, T, A, P) = (\mathbf{1}, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma^5, \gamma^5)$, the Fierz rearrangement used in this paper is the standard linear map [9, 10]

$$F = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 4 & -2 & 0 & 2 & -4 \\ 6 & 0 & -2 & 0 & 6 \\ 4 & 2 & 0 & -2 & -4 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix}, \qquad F^2 = \mathbb{1}. \tag{A1}$$

For the minimal axial–axial contact operator,

$$(J^5 \cdot J^5) \xrightarrow{\text{Fierz}} \tfrac{1}{4}SS + \tfrac{1}{2}VV - \tfrac{1}{2}AA - \tfrac{1}{4}PP. \tag{A2}$$

The coefficients are dimensionless rationals, so rearrangement preserves the common $\kappa = 8\pi/M_{\rm Pl}^2$ prefactor. The matrix, involution, and channel coefficients are checked directly by `fierz_lemma_check.py`.

Equation (A2) is a Dirac-algebra identity, not an assertion that one mean-field factorization is basis-independent. The gap-equation test below chooses the displayed direct-channel convention and reports that choice explicitly. Nor does this identity enumerate derivative operators, curvature–current operators, independent flavor/color contractions, non-minimal torsion irreducible components, or the gravitational EFT.

Appendix B: Regulated Standard Mean-Field NJL Check

For the maximal Einstein–Cartan contact coefficient in Eq. (5), the scalar coefficient selected by Eq. (A2) is

$$G_{\rm scalar} = \left(-\frac{3}{16}\right)\left(+\frac{1}{4}\right)\kappa = -\frac{3}{64}\kappa. \tag{B1}$$

Consider the standard hard-Euclidean-four-momentum-cutoff NJL model with N_f flavors, N_c colors, and scalar contact term $G_s(\bar\psi\psi)^2$. In the chiral limit the mean-field gap equation is

$$M = G_s N_f N_c \frac{M}{2\pi^2}\left[\Lambda^2 - M^2 \ln\!\left(1 + \frac{\Lambda^2}{M^2}\right)\right], \tag{B2}$$

TABLE I. Exact coefficient-to-threshold scan, with $R_S = |G_{\rm scalar}|/G_{\rm crit}$ and R_A the axial Fierz coefficient divided by the *scalar* threshold. Thus R_A is a convention benchmark, not a derived axial-vector condensation threshold. This table tests neither coherent axial order nor a cosmological stress tensor or equation of state.

$N_f N_c$	$\Lambda/M_{\rm Pl}$	R_S	R_A
1	1	0.05968	0.11937
1	$1/\sqrt{0.274}$	0.21782	0.43564
3	1	0.17905	0.35810
3	$1/\sqrt{0.274}$	0.65346	1.30693
9	1	0.53715	1.07430
9	$1/\sqrt{0.274}$	1.96039	3.92079

which gives the bifurcation threshold

$$G_{\rm crit} = \frac{2\pi^2}{N_f N_c \Lambda^2} \tag{B3}$$

in this normalization [11–13]. Here $M = -2G_s\langle\bar\psi\psi\rangle$ and the hard four-dimensional Euclidean ball gives $\langle\bar\psi\psi\rangle = -(N_f N_c M/4\pi^2)[\Lambda^2 - M^2\ln(1+\Lambda^2/M^2)]$, fixing every factor in Eq. (B3). The scalar coefficient in Eq. (B1) is negative and therefore repulsive in this convention. Discarding that sign as a conservative magnitude test gives

$$\frac{|G_{\rm scalar}|}{G_{\rm crit}} = \frac{3N_f N_c}{16\pi}\frac{\Lambda^2}{M_{\rm Pl}^2}, \tag{B4}$$

where the unreduced mass convention $\kappa = 8\pi/M_{\rm Pl}^2$ has been used. The scan fixes $\gamma = 0.274$, takes $N_f N_c = 1, 3, 9$, and evaluates both $\Lambda = M_{\rm Pl}$ and the formal above-Planck sensitivity point $\Lambda = M_{\rm Pl}/\sqrt{0.274}$. The latter is a stress test, not a claim that the contact EFT remains controlled there.

The scalar magnitude is therefore formally supercritical in the last row, and the axial benchmark exceeds unity in three rows. The old blanket magnitude-subcritical statement is false. The scalar-condensate conclusion instead follows from the separate sign result: inserting $G_{\rm scalar} < 0$ into the real homogeneous scalar gap equation admits no nonzero solution. The pseudoscalar coefficient magnitude equals the scalar magnitude.

The result is reproducible with `njl_gap_equation_route1.py`. Its interpretation is strictly standard mean field with this regulator and channel convention. A different Fierz-complete truncation, exchange treatment, multi-species model, or nonperturbative completion can change the mean-field organization, so the calculation is not presented as a regulator- or basis-independent exclusion of every condensate mechanism.

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